Body Control Module (BCM) - High-Level Design

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1 Introduction

This document describes the AUTOSAR high-level software architecture of the Body Control Module (BCM). The BCM manages door locking, exterior lighting and door status monitoring for a passenger vehicle. It runs on a 32-bit microcontroller with AUTOSAR Classic Platform 4.4.

2 System Overview

The BCM receives vehicle speed from the powertrain CAN network and door sensor signals from the door modules. Application software components communicate exclusively through the RTE. The BCM implements auto-lock: all doors are locked automatically when the vehicle speed exceeds 10 km/h. The BCM also switches on the headlamps when the ambient light level falls below the configured threshold.

The design follows the layered AUTOSAR approach. Application SWCs do not access hardware directly; they use the ECU abstraction and complex device drivers.

3 Software Components

3.1 SWC_DoorLockCtrl

Type: Application SWC

Description: Controls central locking. Implements auto-lock above 10 km/h and unlock after a crash event.

Runnables: DoorLockCtrl_Run10ms, DoorLockCtrl_Init

Port	Direction	Interface
pDoorLockCmd	P-Port	If_DoorLockCmd
rVehicleSpeed	R-Port	If_VehicleSpeed
rCrashState	R-Port	If_CrashState
rDoorState	R-Port	If_DoorState

3.2 SWC_VehicleSpeedMgr

Type: Application SWC

Description: Filters and plausibility-checks the wheel pulse signals and provides the vehicle speed in km/h.

Runnables: VehicleSpeedMgr_Run10ms

Port	Direction	Interface
pVehicleSpeed	P-Port	If_VehicleSpeed
rWheelPulse	R-Port	If_WheelPulse

3.3 SWC_WheelSensorIf

Type: Sensor-Actuator SWC

Description: Abstracts the four wheel speed sensors and provides raw pulse counts.

Runnables: WheelSensorIf_Run5ms

Port	Direction	Interface
pWheelPulse	P-Port	If_WheelPulse

3.4 SWC_DoorStateMgr

Type: Application SWC

Description: Aggregates the raw door sensor states into a single door state and reacts to lock commands.

Runnables: DoorStateMgr_Run20ms

Port	Direction	Interface
pDoorState	P-Port	If_DoorState
rDoorSensorRaw	R-Port	If_DoorSensorRaw
rDoorLockCmd	R-Port	If_DoorLockCmd

3.5 SWC_DoorSensorIf

Type: Sensor-Actuator SWC

Description: Reads the door contact sensors through the I/O hardware abstraction.

Runnables: DoorSensorIf_Run20ms

Port	Direction	Interface
pDoorSensorRaw	P-Port	If_DoorSensorRaw

3.6 SWC_LightCtrl

Type: Application SWC

Description: Controls the headlamps based on ambient light and door state. Welcome lighting is activated when a door opens.

Runnables: LightCtrl_Run50ms

Port	Direction	Interface
rDoorState	R-Port	If_DoorState
rIgnitionState	R-Port	If_IgnitionState
pHeadlampCmd	P-Port	If_HeadlampCmd
pHeadlampFeedback	P-Port	If_HeadlampFeedback

3.7 SWC_CrashEventMgr

Type: Application SWC

Description: Evaluates the crash signal from the airbag ECU and provides a debounced crash state.

Runnables: CrashEventMgr_Run10ms

Port	Direction	Interface
pCrashState	P-Port	If_CrashState

4 Port Interfaces

4.1 If_VehicleSpeed

Type: SenderReceiver

Data elements: VehicleSpeed (float32, km/h)

4.2 If_WheelPulse

Type: SenderReceiver

Data elements: PulseCountFL (uint16); PulseCountFR (uint16); PulseCountRL (uint16); PulseCountRR (uint16)

4.3 If_DoorLockCmd

Type: SenderReceiver

Data elements: LockRequest (uint8)

4.4 If_DoorState

Type: SenderReceiver

Data elements: DoorStateFL (uint8); DoorStateFR (uint8); DoorStateRL (uint8); DoorStateRR (uint8)

4.5 If_DoorSensorRaw

Type: SenderReceiver

Data elements: RawContactBits (uint8)

4.6 If_CrashState

Type: SenderReceiver

Data elements: CrashDetected (boolean)

4.7 If_HeadlampCmd

Type: SenderReceiver

Data elements: HeadlampRequest (uint8)

4.8 If_HeadlampFeedback

Type: SenderReceiver

Data elements: LampFault (boolean)

4.9 If_IgnitionState

Type: SenderReceiver

Data elements: IgnitionState (uint8)

5 Timing and Scheduling

Runnables are mapped to periodic OS tasks of 5 ms, 10 ms, 20 ms and 50 ms. The auto-lock decision in DoorLockCtrl_Run10ms shall be made within 20 ms of a vehicle speed update. Data consistency between tasks is ensured by the RTE using implicit communication.

6 Diagnostics

Faults detected by the SWCs are reported to the Diagnostic Event Manager (Dem). Door sensor plausibility failures are stored as DTC 0xB10100. The wheel speed signal loss is stored as DTC 0xC12200.